Task 1

Match the description of each cell organelle.

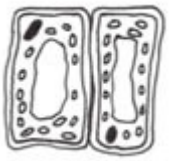
cytoplasm	Place within cells where aerobic respiration takes place to release energy.
nucleus	An organelle in the cell in which protein synthesis takes place.
ribosome	Thin layer around the cell that controls everything that enters and leaves the cell.
mitochondria	Material in a cell but not the material in the nucleus. Most chemical processes take place here, controlled by enzymes.
cell membrane	Filled with cell sap and helps to keep cell turgid depending on the amount of water in cell. Can store water, minerals and even toxins.
cell wall	The control centre of the cell. Contains genetic material.
permanent vacuole	Characteristic of plant cells. Made from cellulose.

Task 2

Match each diagram to the correct cell name.















guard cells	**root hair cell**	**white blood cells**	**palisade cells**
ciliated cells	**sperm cells**	**red blood cells**	

Task 3

Match each specialised cell with its function.

palisade cell	Allows gas exchange in leaves. They control the size of the stomatal pores on the underside of leaves. Controls water loss in leaves.
red blood cell	Adapted so water and mineral ions are easily absorbed from the soil.
sperm cell	Plant cell found in mesophyll layer of leaf. Absorbs light energy for photosynthesis.
white blood cell	Found in respiratory system and moves mucus out of the way.
root hair cell	Contains half the DNA needed to fertilise an egg.
ciliated cell	Engulf bacteria. May release antibodies and antitoxins.
guard cell	Transports oxygen in the blood from lungs to all cells in body.







Task 4

Cut up the statements and match the adaptations with the correct cell. The numbers in brackets show the number of statements for each cell.

palisade cell (4)	white blood cell (2)	sperm cell (3)	neurone (2)
red blood cell (3)	ciliated cell (2)	root hair cell (2)	guard cell (2)

Lots of mitochondria to provide energy for movement.	Shape helps it move smoothly along blood capillaries.
Contains many chloroplasts	Regular and thin shape so they can be packed together.
Adapted to engulf and digest 'foreign' cells.	Produce antibodies to help fight foreign antigens.
No nucleus but lots of room for haemoglobin to which oxygen attaches.	Positioned near top of leaf.
Many cilia that move in unison.	Branched endings to connect with other cells.
Provides a large surface area for the absorption of water.	Long tail and streamlined head to help it swim towards the egg.
Very long to allow nerve impulses to travel to other parts of the body.	Contains many mitochondria to provide cilia with energy.
Concave shape to increase surface area for oxygen absorption.	Shape allows it to grow between soil particles.
Kidney shape with thin outer walls and thick inner walls.	Long shape to provide greater surface area for absorbing carbon dioxide.
Enzymes in head digest through the egg's cell membrane.	Amount of water absorbed affects their shape.

Teaching notes and answers

For the first three tasks students could be asked to cut and paste or simply draw a line to match up the correct statements.

Task 1

cytoplasm	Material in a cell but not the material in the nucleus. Most chemical processes take place here, controlled by enzymes.
nucleus	The control centre of the cell. Contains genetic material.
ribosome	An organelle in the cell in which protein synthesis takes place.
mitochondria	Place within cells where aerobic respiration takes place to release energy.
cell membrane	Thin layer around the cell that controls everything that enters and leaves the cell.
cell wall	Characteristic of plant cells. Made from cellulose.
permanent vacuole	Filled with cell sap and helps to keep cell turgid depending on the amount of water in cell. Can store water, minerals and even toxins.

Task 2

guard cells



ciliated cell



white blood cells



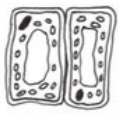
root hair cell



red blood cells



palisade cells



sperm cell



Task 3

palisade cell	Plant cell found in mesophyll layer of leaf. Absorbs light energy for photosynthesis.
red blood cell	Transports oxygen in the blood from lungs to all cells in body.
sperm cell	Contains half the DNA needed to fertilise an egg.
white blood cell	Engulf bacteria. May release antibodies and antitoxins.

root hair cell	Adapted so water and mineral ions are easily absorbed from the soil.
ciliated cell	Found in respiratory system and moves mucus out of the way.
guard cell	Allows gas exchange in leaves. Controls the size of the stoma (pores) underneath leaves. Controls water loss in leaves.

Task 4

palisade cell (4)	Contains many chloroplasts. Positioned near top of leaf. Regular and thin shape so they can be packed together. Long shape to provide greater surface area for absorbing carbon dioxide.
red blood cell (3)	No nucleus but lots of room for haemoglobin to which oxygen attaches. Concave shape to increase surface area for oxygen absorption. Shape helps it move smoothly along blood capillaries.
sperm cell (3)	Long tail and streamlined head to help it swim towards the egg. Lots of mitochondria to provide energy for movement. Enzymes in head digest through the egg's cell membrane.
white blood cell (2)	Adapted to engulf and digest 'foreign' cells. Produce antibodies to help fight foreign antigens.
root hair cell (2)	Provides a large surface area for the absorption of water. Shape allows it to grow between soil particles.
ciliated cell (2)	Many cilia that move in unison. Contains many mitochondria to provide cilia with energy.
guard cell (2)	Kidney shape with thin outer walls and thick inner walls. Amount of water absorbed affects their shape.
neurone (2)	Branched endings to connect with other cells. Very long to allow nerve impulses to travel to other parts of the body.